

SAI SAKETH BOLLA

[Open to Relocation] 📍 College Park, MD 📞 +1 (240) 423-9350

✉️ saisakethbollamr@gmail.com 🌐 linkedin

WORK EXPERIENCE

Operations & Software Engineer (Part time)

April 2025 – Dec 2025

Univeristy of Maryland (Robust Quantum Simulation)

College Park, MD

- Redesigned and rebuilt the RQS website using **React, Next.js, and Node.js**, replacing a static CMS with a dynamic, version-controlled platform that reduced update time from hours to minutes.
- Developed backend APIs in **Python (FastAPI)** to manage content, authentication, and analytics, improving reliability and maintainability of the platform.
- Integrated CI/CD pipelines with **GitHub Actions** for automated testing and deployment, ensuring rapid and error-free releases.
- Implemented logging, monitoring, and alerting tools to track web performance and user activity, providing insights that guided feature improvements.

Software Engineer

Feb 2024 – Dec 2024

Prodapt Solutions

Hyderabad, India

- Developed **centralized log analysis system** processing **50M+ daily telecom network logs** across 200+ devices for critical infrastructure monitoring.
- Engineered custom search indexing using **PostgreSQL and Python hash maps**, reducing log query time from 5+ minutes to **2 seconds (150× performance improvement)**.
- Implemented automated storage optimization with **compression algorithms and TTL policies**, achieving **45% cost reduction (\$45K annual savings)** while maintaining full data accessibility.
- Designed and deployed scalable data pipeline using **Python collectors, Flask REST APIs, and React dashboard**, enabling real-time network troubleshooting.
- Delivered production system to **2 major telecom clients**, reducing average incident resolution time from 8 hours to **1 hour** and generating **\$1.2M in company revenue**.

Software Engineer

May 2022 – Jan 2024

Verizon (Contract through Prodapt)

Hyderabad, India

- Designed and implemented large-scale data infrastructure using **Python and SQL**, supporting critical business operations for 30,000+ users across Verizon's enterprise systems.
- Built high-throughput ETL platform processing 100+ TB data migration from MySQL to **Google BigQuery**, achieving 40% query performance improvement through optimized data modeling and partitioning strategies.
- Engineered workflow orchestration system using **Apache Airflow on Google Cloud Composer**, implementing event-driven architecture with GCS triggers and automated failure recovery mechanisms.
- Optimized distributed query processing by migrating HiveQL workloads to **BigQuery SQL**, leveraging columnar storage and query optimization techniques for improved scalability.
- Implemented data quality assurance framework with synthetic data generators and automated validation pipelines ensuring data integrity across petabyte-scale migrations.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Rust (Beginner)

Web & Application Development: Flask, FastAPI, React, Node.js, REST APIs, Object-Oriented Design

Databases: PostgreSQL, MySQL, Google BigQuery

Data & Distributed Systems: Apache Spark, Kafka Airflow, Data Pipelines, SQL Optimization

Cloud & DevOps (working knowledge): AWS (EC2, Lambda, CloudFront, S3), GCP (GKE, BigQuery), Kubernetes, Docker, Terraform

Testing & CI/CD: GitHub Actions, Jenkins, PyTest, Unit/Integration Testing

Other: Linux, Git, Agile Development

Certifications: Google Cloud Associate + Professional

EDUCATION

University of Maryland, College Park, USA

Jan 2025 – Dec 2026

Master of Engineering in Computer Software Engineering

College Park, MD

Coursework: Data Structures and Algorithms, Software Engineering, Linux System Admin

PROJECTS

AI-Organized File Management | PowerShell, Windows Explorer, OpenAI (GPT-4 Mini)

- Built **FileSense**, a PowerShell tool that leverages **OpenAI GPT-4 Mini** to safely auto-sort files from Windows Explorer.
- Implemented **batch processing, undo system, and configurable alerts** to improve reliability, efficiency, and control over large folders.

Stream Command Output to Webhooks | Rust, CLI, Slack API, Google Chat API

- Built **PipeBeam**, a Rust CLI for **real-time stdout/stderr streaming** to Slack and Google Chat, enhancing script and job observability.
- Added **smart buffering, interactive shell mode, and custom alerts** to improve reliability and reduce noisy output.

ACHIEVEMENTS

- 4th rank in International E-school Flipkart grid warehouse robot making competition 2022.